Experiment 9 — Python Modules and Standard Libraries

Python Lab Manual — Program and Output

EX. NO: 9(a) — Pandas

PROGRAM

```
import pandas as pd

df = pd.DataFrame({
    "Name": ["Mr. Owen Harris", "Mr. William Henry", "Miss. Elizabeth"],
    "Age": [22, 35, 58],
    "Sex": ["male", "male", "female"]
})

print(df)
print(df["Age"])

ages = pd.Series([22, 35, 58], name="Age")
print(ages)
print(ages.max())
print(df.describe())
```

OUTPUT / RESULT

```
The supplied PDF shows a 3-row DataFrame, the Age Series (22, 35, 58),
maximum age 58, and descriptive statistics.
Note: the sample names in the printed output in the source PDF differ from
the names in its program.
```

EX. NO: 9(b) — NumPy

PROGRAM

```
import numpy as np

a = np.arange(6)
a2 = a[np.newaxis, :]
print(a2.shape)

# Array creation and functions
a = np.array([[1, 2, 3, 4],
              [5, 6, 7, 8],
              [9, 10, 11, 12]])
print(a[0])
print(a[1])

print(np.zeros(2))
print(np.ones(2))
print(np.arange(4))
print(np.arange(2, 9, 2))
print(np.linspace(0, 10, num=5))

x = np.ones(2, dtype=np.int64)
print(x)

arr = np.array([2, 1, 5, 3, 7, 4, 6, 8])
print(np.sort(arr))

a = np.array([1, 2, 3, 4])
b = np.array([5, 6, 7, 8])
print(np.concatenate((a, b)))

array_example = np.array([
    [[0, 1, 2, 3], [4, 5, 6, 7]],
    [[0, 1, 2, 3], [4, 5, 6, 7]],
    [[0, 1, 2, 3], [4, 5, 6, 7]]
])
print(array_example.ndim)
print(array_example.size)
print(array_example.shape)

a = np.arange(6)
print(a)
b = a.reshape(3, 2)
print(b)
```

```
print(np.reshape(a, newshape=(1, 6), order="C"))
```

OUTPUT / RESULT

```
Selected outputs shown in the supplied PDF:
[1 2 3 4]
[5 6 7 8]
[1 1]
[0 1 2 3 4 5]
[[0 1]
 [2 3]
 [4 5]]
array([[0, 1, 2, 3, 4, 5]])
```

EX. NO: 9(c) — Matplotlib pie plot

PROGRAM

```
import matplotlib.pyplot as plt

slices = [7, 2, 2, 13]
activities = ["sleeping", "eating", "working", "playing"]
cols = ["c", "m", "r", "b"]

plt.pie(
    slices,
    labels=activities,
    colors=cols,
    startangle=90,
    shadow=True,
    explode=(0, 0.1, 0, 0),
    autopct="%1.1f%%"
)
plt.title("Pie Plot")
plt.show()
```

OUTPUT / RESULT

```
Output: A pie chart titled "Pie Plot" is displayed.
```

EX. NO: 9(d) — SciPy interpolation

PROGRAM

```
import matplotlib.pyplot as plt
from scipy import interpolate
import numpy as np

x = np.arange(5, 20)
y = np.exp(x / 3.0)
f = interpolate.interp1d(x, y)

x1 = np.arange(6, 12)
y1 = f(x1)

plt.plot(x, y, "o", x1, y1, "--")
plt.show()
```

OUTPUT / RESULT

```
Output: A plot showing the original points and interpolated values is displayed.
```